GPT Prompt: Luthier's Rosette Ring SVG with Geometric Tile Extensions

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MASTER PROMPT — COPY EVERYTHING BELOW THIS LINE

You are a precision SVG engineering assistant with expertise in geometric drafting, luthier instrument construction, and CAD illustration. Your task is to generate a complete, accurate, well-structured SVG file depicting a **guitar rosette ring layout** as seen in a professional CAD drawing, including all concentric ring channels, radial segment divisions, geometric tile extensions on the outer edges, and technical dimension annotations.

DOCUMENT SPECIFICATIONS

- **SVG canvas:** 900 × 900 px
- **Viewbox:** `0 0 900 900`
- **Center point (CX, CY):** 450, 450
- **Background:** `#f5f0e8` (warm drafting paper tone)
- **Stroke color (default):** `#1a1a2e` (near-black technical ink)
- **All geometry must be mathematically precise** using SVG path arcs, not approximations
- **Include a title block** in the lower-right corner with: Title, Scale, Units, Date fields

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PART 1 — THE ROSETTE RING (Primary Drawing, Left/Center)

1A. Overall Ring Structure

Draw a **full circular rosette** consisting of **5 concentric annular ring zones**, each representing a distinct inlay channel in a guitar soundhole rosette. All measurements are in **inches**, with `1 inch = 100 SVG units` for this drawing.

Ring Zone Table (from innermost to outermost):

Zone #	Name	Inner Radius (r_1)	Outer Radius (r_2)	Radial Depth	Fill Color
1	Soundhole Binding	1.50" (150 px)	1.90" (190 px)	0.40"	#d4b483 (maple cream)
2	Inner Purfling Channel	1.90" (190 px)	2.10" (210 px)	0.20"	#c8c8c8 (silver/abalone suggestion)
3	Main Rosette Channel	2.10" (210 px)	3.00" (300 px)	0.90"	**Tiled — see Part 1C**
4	Outer Purfling Channel	3.00" (300 px)	3.20" (320 px)	0.20"	#c8c8c8
5	Outer Binding	3.20" (320 px)	3.50" (350 px)	0.30"	#d4b483

- Draw each ring as a full closed annular shape using SVG `<path>` with arc commands
- Ring boundaries must be drawn as thin precise circles: `stroke="#1a1a2e"` `stroke-width="0.8"` `fill="none"` overlaid on the fills
- The **center soundhole** (r < 1.50" / 150 px) should be filled #2a1f0e (dark void/ebony)
- Label each ring zone with a radial callout leader line pointing to the midpoint arc of the ring, with the zone name in `font-size="9"` `font-family="monospace"`

1B. Radial Segment Divisions

Divide the entire rosette (all ring zones) into **12 equal radial segments** at **30° intervals**, beginning at 0° (top / 12 o'clock position) and rotating clockwise.

- Draw each radial spoke as a line from inner soundhole boundary (r = 150 px) to outer binding edge (r = 350 px)
- Spoke style: `stroke="#1a1a2e"` `stroke-width="0.6"` `stroke-dasharray="none"`
- Add a **30.0°** angle annotation arc between spoke 0 and spoke 1, shown as a small arc near r = 380 px with a leader and "30.0°" label in `font-size="8"`
- The angular position of each spoke shall be: `angle_n = n × 30°`, where n = 0 through 11

1C. Main Rosette Channel Tile Fill (Zone 3 — the primary inlay band)

The Main Rosette Channel (r_1=210px, r_2=300px, depth=0.90") is where the decorative geometric tile motif lives. Fill this zone using **alternating geometric tile segments** as follows:

Tile pattern — Herringbone / Alternating Trapezoid:

- Alternate between two fill colors segment by segment:
  - **Even segments (0,2,4,6,8,10):** `#4a90d9` (blue — represents abalone/shell inlay)
  - **Odd segments (1,3,5,7,9,11):** `#b0c8e8` (light blue — represents lighter material or negative space)
- Each tile segment is an **arc-bounded trapezoid** defined by:
  - Inner arc: radius 210 px, spanning 30°
  - Outer arc: radius 300 px, spanning 30°
  - Two straight radial edges connecting inner to outer at each spoke
- Each tile path uses `<path>` with SVG arc commands: `M`, `L`, `A`
- Add a subtle diagonal hatch stroke pattern inside each tile to suggest wood grain or inlay texture (use `<pattern>` defs or simply draw 3–4 diagonal lines per cell at 45°, `opacity="0.25"`)

1D. Geometric Extensions on Ring Edges (CRITICAL FEATURE)

This is the defining visual feature of the drawing. Along **both the inner edge (r=210px) and the outer edge (r=300px)** of the Main Rosette Channel, draw **geometric tile tab extensions** — trapezoidal protrusions that project outward/inward from the ring boundary, mimicking the physical tile geometry used in actual rosette construction.

Extension geometry rules:

- At the midpoint of each 30° segment, draw a **trapezoidal tab** that projects:
  - **Outward** (into Zone 4) by 12 px from r=300 to r=312
  - **Inward** (into Zone 2) by 10 px from r=210 to r=200
- Each tab is centered on its segment's angular midpoint (at 15°, 45°, 75°, 105°, 135°, 165°, 195°, 225°, 255°, 285°, 315°, 345°)
- Tab shape: a **parallelogram** with angular width of 10° at the ring boundary, tapering or maintaining width — matching the tile's trapezoidal geometry
- Fill tabs with the same color as their parent segment tile
- Stroke tabs with `stroke="#1a1a2e"` `stroke-width="0.8"`
- These tabs represent the physical **tongue-and-groove** or **key register** geometry that locks inlay tiles into the channel during construction

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PART 2 — DETAIL VIEW: INDIVIDUAL TILE PIECE (Upper Right)

Draw an **orthographic 3-view detail** of a single tile piece in the upper-right quadrant of the SVG

(approximately at x=620, y=80, within a 240×200 px region).

2A. Tile Shape Description

The tile is a **trapezoidal parallelogram** — the physical piece a luthier would cut from wood, shell, or purfling strip. It represents one segment of the main rosette channel.

Tile dimensions (at 1:1 scale, 1 inch = 100 SVG units):

- **Overall width (wide end):** 1.50" (150 px)
- **Overall height (depth of channel):** 2.000" (200 px) — this is the radial depth rendered vertically for clarity in the detail view
- **Side offset (parallelogram skew):** 0.75" (75 px) on each side — the piece is skewed because it must fit the arc curvature of the ring
- **Narrow end:** 0.625" (62.5 px) — the inner arc edge is shorter than the outer arc edge
- **Side walls:** angled, not perpendicular — they represent the radial spoke cuts

Rendered as:

- A 4-point polygon / `<polygon>` or `<path>` showing the trapezoidal profile
- Fill: `#4a90d9` with a diagonal stripe texture (`opacity="0.3"`)
- Stroke: `#1a1a2e` `stroke-width="1.2"`

2B. Dimension Annotations for the Detail View

Add fully dimensioned engineering callouts in standard orthographic drafting style:

Dimension	Value	Location
Overall width	1.50"	Top horizontal leader with arrows
Channel depth	2.000"	Left vertical leader with arrows
Left offset	0.75"	Bottom partial leader
Right offset	0.75"	Bottom partial leader
Narrow end	0.625"	Right side vertical, small
Skew angle	~14°–18° (computed from offsets)	Arc annotation on corner

- All dimension lines: `stroke="#333"` `stroke-width="0.7"` with arrowheads (small filled triangles, 4px)
- All dimension text: `font-family="monospace"` `font-size="9"` `fill="#1a1a2e"`

- Extension lines extend 4px beyond the geometry before meeting the dimension line
- Dimension lines are offset 18px from the geometry edge

2C. Isometric / Perspective Thumbnail (Lower Right of detail region)

Below the orthographic detail (approximately x=660, y=290), draw a small **isometric projection** of the same tile piece to give 3D context:

- Show the tile as a flat slab approximately 0.08" thick (8 px in this scale)
- Apply a top face fill of `#4a90d9`, a left face fill of `#2a5a90`, a right face fill of `#3a70b0`
- Stroke all edges `#1a1a2e` `stroke-width="0.8"`
- Label: "ISO VIEW" in `font-size="7"` below the shape

PART 3 — DIMENSION ANNOTATIONS ON THE RING

Add the following engineering dimension callouts to the primary ring drawing:

Annotation	Value	Placement
Overall outer diameter	Ø 7.00" (Ø700px)	Horizontal spanning dimension above the ring, at y = CY − 370px, with arrows touching each outer edge
Main channel outer diameter	Ø 6.00" (Ø600px)	Horizontal dimension line at y = CY − 320px
Inner soundhole diameter	Ø 3.00" (Ø300px)	Horizontal dimension inside or below the soundhole
Ring zone radial depths	per Ring Zone Table	Radial callout leaders pointing to each ring, right side
Segment angle	30.0°	Arc annotation between segment 0 and segment 1, near outer edge
Circumferential note	"12 EQUAL DIVISIONS × 30°"	Text label arced or straight, below the ring at CY + 370px, centered

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PART 4 — SUPPORTING GRAPHIC ELEMENTS

4A. Center Registration Marks

- Draw a **center crosshair** at (450, 450): horizontal and vertical lines each 30 px long, `stroke-width="0.8"`

`stroke="#1a1a2e"` `stroke-dasharray="2,2"`

- Draw a small center circle: `r="4"` `fill="none"` `stroke="#1a1a2e"` `stroke-width="0.8"`

4B. Section Line Indicators

- Draw a horizontal section cut line (A–A) across the full ring diameter, `stroke-dasharray="8,4,2,4"` `stroke="#1a1a2e"` `stroke-width="0.8"`
- Label both ends: "A" in a small circle `r="8"` `fill="#1a1a2e"` with white text

4C. Title Block (Lower Right Corner)

At approximately x=620, y=760, draw a simple bordered title block containing:

```
LUTHIER'S ROSETTE LAYOUT
Scale: 1:1 | Units: Inches
Segments: 12 | Angle: 30° ea.
Main Channel OD: Ø6.00"
Drawing: ROSETTE-001
```

- Box style: `stroke="#1a1a2e"` `stroke-width="1"` `fill="white"` `fill-opacity="0.85"`
- Text: `font-family="monospace"` `font-size="8"` `fill="#1a1a2e"`
- Internal horizontal dividers between each line

CONSTRUCTION NOTES (Embed as SVG `<desc>` or comment block)

Include this text as an SVG comment block:

```
<!--
 ROSETTE CONSTRUCTION NOTES:
 - All 12 tile pieces are geometrically identical (rotational symmetry)
 - Tile skew angle compensates for arc curvature of main channel
 - Wide end (1.50") seats at outer boundary r=3.00"
 - Narrow end (0.625") seats at inner boundary r=2.10"
 - 0.75" offsets define the parallelogram skew for tangential alignment
 - Physical tile thickness: approximately 0.080" (2mm)
 - Material options: herringbone purfling strip, abalone shell, MOP, figured maple
 - Extension tabs register tile position during glue-up
 - Channel routed to depth = tile thickness + 0.005" clearance
-->
```

OUTPUT REQUIREMENTS

- Output **only** the complete SVG file, beginning with `<svg` and ending with `</svg>`
- Use `<defs>` for any reusable patterns, markers (arrowheads), and gradients
- Group related elements with `<g id="...">` tags:
  - `#ring-background` — filled ring zones
  - `#ring-cells` — individual tile segments
  - `#ring-extensions` — geometric tab extensions
  - `#ring-guides` — spokes, boundary circles, crosshairs
  - `#ring-dimensions` — all dimension annotations
  - `#detail-view` — tile detail orthographic drawing
  - `#title-block` — title block
- **No external fonts, no external images, no JavaScript**
- All geometry must be mathematically computed — no hand-approximated coordinates
- The SVG must render correctly in any standards-compliant SVG viewer (Inkscape, browser, Illustrator)

END OF PROMPT